Supplementary material

Phase Dynamics, Recurrence, and Relational Temporal Ordering in SEFI-DEFI-GWFM

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S1. Model and reduction conventions

The main article analyzes the base DEFI identity action with positive constant metric coefficients and a common unit frequency. Geometry is prescribed, not varied. A linear coupling example can be realized with $\alpha = \beta = \gamma = \delta = 1$, $g_1 = 1$, $\kappa_0^2 = r$, and $g_2 = 0$. General normalized directions give the same constant-geometry spectrum.

The effective potential is an algebraic completion of squares, not an elimination of dynamical coordinates. Its coercivity theorem is a sufficient global condition. A positive potential Hessian yields local nonlinear stability; a zero Hessian eigenvalue needs higher-order analysis. The article does not equate these different stability statements.

S2. Figure parameters and exact examples

Figure 1 evaluates the eigenvalue branches $1 - r, 1, 1 + r$ on $0 \le r \le 1.5$. Figure 2 uses $\eta_- = \cos(\sqrt{1/2}\,\tau)$, $\eta_- = \tau$, and $\eta_- = \exp(\sqrt{1/5}\,\tau)$, respectively, through $\tau = 12$. Initial states intentionally differ as stated in the caption. Origin is exactly $I_O = \cos\tau$, with $\alpha = 1$ and $Q_O = 1/2$.

Figure 3 evaluates linear effective potentials $\beta(1 - r^2)I_A^2/2$ with $\beta = 1$ and $r = 0.5, 1.2$, and the nonlinear example $I_A^2/2 - I_A^4/8$. The saddle points are at $\pm\sqrt{2}$, with barrier energy $1/2$.

Figure 4 uses $r_0 = 0.5$, $\varepsilon = 0.1$, fixed direction $b = 0$, and drives $\Omega = \sqrt{2}, 2.2$. The same initial state is used for both responses. For the resonant drive it is a unit Euclidean-norm eigenvector of the growing one-period multiplier:

$$(\eta_-(0), \dot{\eta}_-(0)) = (0.8189831212557845, 0.5738176078669363).$$

Both responses are integrated with step 0.01 through $\tau = 120$. Panel (b) shows the Euclidean phase-space norm on a logarithmic axis, not a probability, fidelity, or measured energy. All quantities are dimensionless.

S3. Fundamental matrix and convergence

For $\Omega = \sqrt{2}$, one period is $P = 2\pi/\sqrt{2}$. A fourth-order Runge-Kutta fundamental-matrix integration begins at the two-dimensional identity matrix. At 1,024 steps per period its entries are

$$\mathcal{M}_{11} = \mathcal{M}_{22} = -1.01233345770787,$$

$$\mathcal{M}_{12} = -0.224850304020537, \quad \mathcal{M}_{21} = -0.110380235876938.$$

The corresponding multipliers are -0.8547928927895374 and -1.1698740226262063, with growth rate 0.03531402295026381. The trace of the differential equation matrix is zero, so the Wronskian and monodromy determinant are exactly one analytically.

Resonant drive: successive maximum matrix-entry differences for 128/256, 256/512, and 512/1,024 steps are 1.6927e-08, 1.0567e-09, 6.5998e-11. Observed orders are 4.00172, 4.00095.

Comparison drive: successive maximum matrix-entry differences for 128/256, 256/512, and 512/1,024 steps are 1.5084e-09, 9.4873e-11, 5.949e-12. Observed orders are 3.99085, 3.99527.

The comparison multipliers are [[-0.43699428291805187, 0.899464283169118], [-0.43699428291805187, -0.899464283169118]]. Their distinct unit-modulus pair is consistent with bounded Floquet evolution. This stability statement is well separated from the repeated-multiplier boundary.

Propagating the resonant Floquet initial vector through 27 periods agrees with repeated multiplication by the one-period multiplier to maximum relative error 8.86912277777435e-15. Halving the direct response step from 0.01 to 0.005 gives maximum relative state discrepancy 5.773584549912017e-10 at the recorded sample points. These are deterministic discretization checks, not statistical confidence intervals or physical validation.

S4. Static checks and recurrence

Random matrix tests use 1,000 samples for the constant linear spectrum and 1,000 generalized equilibrium Hessians. Maximum absolute eigenvalue discrepancies are below 9×10^{-16} and 1.5×10^{-15}, respectively. Independent integration at $r = 0.5, 1, 1.2$ with steps $0.1, 0.05, 0.025$ gives approximately fourth-order convergence through $\tau = 8$. Finest Origin-energy drift is below 2×10^{-9}.

At $r = 24/25$, every excited linear frequency satisfies $\omega_j(10\pi) \in 2\pi\mathbb{Z}$, providing exact full-state recurrence analytically. Evaluating the complex phases numerically gives return error below 2.5×10^{-15}. This check does not apply to arbitrary nonlinear or periodically driven dynamics.

S5. DNA extension and helical arithmetic

The DNA comparison adopts $U_{\text{pen}} = \lambda \Delta^2 I^T M I$. The baseline and modified Origin-energy derivatives are obtained by direct differentiation and substitution of the modified equation. For constant mismatch only the modified energy is conserved; for zero mismatch the base action is recovered. This distinction depends on inclusion of Origin in the norm.

The rounded geometric values give $21\pi/34 = 1.9403954625113429$ and $\pi/\varphi = 1.9416110387254664$, a relative discrepancy of approximately 0.000626. Approximate proximity is not exact locking, and a geometric ratio is not by itself a topological invariant.

S6. Reproducibility and data

The accompanying numerical scripts compute the driven response and validation summaries using fourth-order Runge-Kutta integration and NumPy linear algebra. CSV columns identify the evolution parameter, prescribed coupling, and response coordinates and velocities. JSON files record step sizes, convergence differences, and initial data. Analytic figures are specified by the formulas in S2. No experiments or parameter fits were conducted.